

Appendix: Signals from Abroad: Climate Policy Appropriateness and Informational Spillovers

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A Formal Proofs

Effects of Domestic Climate Action

Proof of Proposition 1: The proof of the proposition proceeds in three parts. First I show existence and uniqueness of the equilibrium in Lemma A.1. The first bullet of the proposition is given by Lemma A.2 and the second and third bullets are given by Lemma A.3.

Lemma A.1 *Let $\mathcal{I} = \{a_i\}$. There exists a unique equilibrium.*

Proof of Lemma A.1: The voter observes $a \in \{0, 1\}$ and so his posterior belief about politician appropriateness is $\mu(a) = P(\omega = a|a)$. He reelects the politician iff $\mu(a) \geq \varepsilon$, and so the probability that the politician is reelected is simply $F(\mu(a))$.

Given her signal x , the politician does not know the state with certainty but has the updated posterior belief $\eta(x) = P(\omega = 1|x) = \frac{\pi g(x;1)}{\pi g(x;1) + (1-\pi)g(x;0)}$ where $g(x;\omega)$ is the density of the politician's signal. Then it is optimal for the politician to choose $a = 1$ iff

$$\eta(x) + F(\mu(1)) \geq 1 - \eta(x) + F(\mu(0)).$$

Define $\tilde{x}$ as the signal that makes the politician indifferent between choosing $a = 1$ and $a = 0$. Then with probability $1 - G(\tilde{x}; \omega)$, the politician chooses $a = 1$ in state ω . Given this cutoff, the voter's posterior beliefs can be written as

$$\begin{aligned} \mu(1; \tilde{x}) &= \frac{\pi(1 - G(\tilde{x}; 1))}{\pi(1 - G(\tilde{x}; 1)) + (1 - \pi)(1 - G(\tilde{x}; 0))} \\ \mu(0; \tilde{x}) &= \frac{(1 - \pi)G(\tilde{x}; 0)}{(1 - \pi)G(\tilde{x}; 0) + \pi G(\tilde{x}; 1)}. \end{aligned}$$

The politician's problem can be rewritten such that the equilibrium cutoff signal $\tilde{x}^*$ satisfies

$$2\eta(\tilde{x}^*) - 1 + F(\mu(1; \tilde{x}^*)) - F(\mu(0; \tilde{x}^*)) = 0. \quad (1)$$

By the MLRP, $\eta(x)$ is increasing in x . Observe that $\mu(1; \tilde{x})$ is increasing in $\tilde{x}$ and $\mu(0; \tilde{x})$ is decreasing in $\tilde{x}$, because the MLRP implies the hazard rate order. Hence the LHS is strictly increasing in $\tilde{x}$ such that the politician chooses $a = 1$ iff $x \geq \tilde{x}^*$, which is unique and well-defined. As $\tilde{x}^*$ is unique, the equilibrium is unique. ■

Lemma A.2 *The equilibrium probability of action is increasing in π .*

Proof of Lemma A.2: Define $I(\tilde{x})$ as

$$I(\tilde{x}) = 2\eta(\tilde{x}) - 1 + F(\mu(1; \tilde{x})) - F(\mu(0; \tilde{x})).$$

This function is increasing in $\tilde{x}$ by definition of the equilibrium, proved in Lemma A.1. By the implicit function theorem,

$$\frac{d\tilde{x}^*}{d\pi} = -\frac{\partial I(\tilde{x})/\partial \pi}{\partial I(\tilde{x})/\partial \tilde{x}}.$$

Observe that as $\pi \rightarrow 0$, $I(\tilde{x}) < 0$ so $\tilde{x}^* \rightarrow \infty$; as $\pi \rightarrow 1$, $I(\tilde{x}) > 0$ so $\tilde{x}^* \rightarrow -\infty$. By the MLRP, $\eta(\tilde{x})$ is increasing in π . Moreover, $\mu(1; \tilde{x})$ is increasing in π and $\mu(0; \tilde{x})$ is decreasing in π . Hence $I(\tilde{x})$ is increasing in π and $\frac{d\tilde{x}^*}{d\pi} \leq 0$.

The equilibrium probability of action is then

$$A(\tilde{x}^*) = \pi(1 - G(\tilde{x}^*; 1)) + (1 - \pi)(1 - G(\tilde{x}^*; 0)).$$

Differentiating with respect to π yields

$$\begin{aligned} \frac{dA(\tilde{x}^*)}{d\pi} &= 1 - G(\tilde{x}^*; 1) - \pi g(\tilde{x}^*; 1) \frac{d\tilde{x}^*}{d\pi} - 1 + G(\tilde{x}^*; 0) - (1 - \pi)g(\tilde{x}^*; 0) \frac{d\tilde{x}^*}{d\pi} \\ &= G(\tilde{x}^*; 0) - G(\tilde{x}^*; 1) - \frac{d\tilde{x}^*}{d\pi} (\pi g(\tilde{x}^*; 1) + (1 - \pi)g(\tilde{x}^*; 0)) > 0, \end{aligned}$$

where the result follows from first-order stochastic dominance. ■

Lemma A.3 *There exists a unique $\pi^\dagger$ such that $\mu(1) \geq \mu(0)$ iff $\pi \geq \pi^\dagger$.*

Proof of Lemma A.3: Clearly $F(\mu(1)) \geq F(\mu(0))$ iff $\mu(1) \geq \mu(0)$ so there are positive electoral returns to choosing $a = 1$. In equilibrium, $\mu(1) = \mu(0)$ iff

$$\begin{aligned} \frac{\pi(1 - G(\tilde{x}; 1))}{\pi(1 - G(\tilde{x}; 1)) + (1 - \pi)(1 - G(\tilde{x}; 0))} &= \frac{(1 - \pi)G(\tilde{x}; 0)}{(1 - \pi)G(\tilde{x}; 0) + \pi G(\tilde{x}; 1)} \\ \Leftrightarrow \pi^2 G(\tilde{x}^*; 1)(1 - G(\tilde{x}^*; 1)) &= (1 - \pi)^2 G(\tilde{x}^*; 0)(1 - G(\tilde{x}^*; 0)). \end{aligned}$$

Call any π that satisfies this equality $\pi^\dagger$. Now observe that at $\pi = \pi^\dagger$, $\mu(1) = \mu(0)$ and so the politician's optimal threshold solves

$$2\eta(x) - 1 = 0,$$

which has a unique solution because $\eta(x)$ is increasing in both π and x by the MLRP. So $\pi^\dagger$ must be unique.

Note that $\lim_{\pi \rightarrow 0} \mu(0) = 1$ and $\lim_{\pi \rightarrow 1} \mu(1) = 1$. So for any nonconstant μ on π these posterior beliefs must cross.

By state symmetry of $g(\cdot; \omega)$, we can further characterize $\pi^\dagger = \frac{1}{2}$. At $\pi = \pi^\dagger$, the politician's threshold solves $2\eta(x) - 1 = 0$ or, using the definition of $\eta(x)$, $\frac{\pi^\dagger g(x; 1)}{\pi^\dagger g(x; 1) + (1 - \pi^\dagger)g(x; 0)} = \frac{1}{2}$. Rearranging gives $\frac{g(x; 1)}{g(x; 0)} = \frac{1 - \pi^\dagger}{\pi^\dagger}$. By definition of state symmetry, $\frac{g(c; 1)}{g(c; 0)} = 1$ which is satisfied iff $\pi^\dagger = \frac{1}{2}$. Substituting $\pi^\dagger = \frac{1}{2}$ into the indifference between $\mu(1) = \mu(0)$ gives

$$G(x; 1)(1 - G(x; 1)) = G(x; 0)(1 - G(x; 0)),$$

which holds iff $x = c$. ■ ■

Effects of Special Interest Messaging

Proof of Proposition 2: The proof of the proposition proceeds in four parts. First I show existence and uniqueness of the equilibrium in Lemma A.4. The second bullet of the proposition is given by Lemma A.5. The first bullet of the proposition is given by Lemma A.6. The third bullet of the proposition is given by Lemma A.7.

Lemma A.4 *Let $\mathcal{I} = \{a_i, s\}$. There exists a unique equilibrium.*

Proof of Lemma A.4: The voter now conditions his reelection decision on both a and s such that $\mu(a, s) = P(\omega = a|a, s)$, and reelects the incumbent politician iff $\mu(a, s) \geq \varepsilon$. The incumbent's probability of reelection is simply $F(\mu(a, s))$.

Importantly the politician does not know which signal will be realized, but can update her beliefs about the likelihood of s given her own signal x , giving posterior beliefs $\eta(x) = P(\omega = 1|x) = \frac{\pi g(x;1)}{\pi g(x;1) + (1-\pi)g(x;0)}$. The expected utility from choosing $a = 1$ vs. $a = 0$ can be expressed as

$$\eta(x) + \eta(x)(1-\beta)F(\mu(1,1)) + (1-\eta(x) + \eta(x)\beta)F(\mu(1,0)) \geq 1 - \eta(x) + \eta(x)(1-\beta)F(\mu(0,1)) + (1-\eta(x) + \eta(x)\beta)F(\mu(0,0)).$$

Define $\Delta(s) = F(\mu(1, s)) - F(\mu(0, s))$ so that the politician is indifferent between $a = 1$ and $a = 0$ when

$$2\eta(\tilde{x}) - 1 + \eta(\tilde{x})(1-\beta)\Delta(1) + (1-\eta(\tilde{x}) + \eta(\tilde{x})\beta)\Delta(0) = 0, \quad (2)$$

where $\tilde{x}$ is the signal that satisfies at equality, as before. Given this cutoff, voters have posterior beliefs

$$\begin{aligned} \mu(1, 1; \tilde{x}) &= 1. \\ \mu(1, 0; \tilde{x}) &= \frac{\pi\beta(1 - G(\tilde{x}; 1))}{\pi\beta(1 - G(\tilde{x}; 1)) + (1 - \pi)(1 - G(\tilde{x}; 0))}. \\ \mu(0, 0; \tilde{x}) &= \frac{(1 - \pi)G(\tilde{x}; 0)}{(1 - \pi)G(\tilde{x}; 0) + \pi\beta G(\tilde{x}; 1)}. \\ \mu(0, 1; \tilde{x}) &= 0. \end{aligned}$$

Differentiating Equation 2 with respect to $\tilde{x}$ yields

$$2\frac{\partial\eta(\tilde{x})}{\partial\tilde{x}} + (1-\beta)\frac{\partial\eta(\tilde{x})}{\partial\tilde{x}}(\Delta(1; \tilde{x}) - \Delta(0; \tilde{x})) + (1-\eta(\tilde{x})\beta)\eta(\tilde{x})\frac{\partial\Delta(0; \tilde{x})}{\partial\tilde{x}}.$$

Since $g(\cdot)$ has the monotone likelihood ratio property, $\eta(\tilde{x})$ is increasing in $\tilde{x}$. Now observe that $\mu^*(1, 0; \tilde{x})$ is increasing in $\tilde{x}$ and $\mu^*(0, 0; \tilde{x})$ is decreasing in $\tilde{x}$, which means that $\Delta(0)$ is increasing in $\tilde{x}$. Further, by definition of posterior beliefs we have $\Delta(1; \tilde{x}) \geq \Delta(0; \tilde{x})$ so this expression is increasing in $\tilde{x}$. Hence by the intermediate value theorem there is a unique $\tilde{x}^*$ solving Equation 2 such that the politician chooses $a = 1$ iff $x \geq \tilde{x}^*$. ■

Lemma A.5 *The equilibrium is identical to that without the interest group as $\beta \rightarrow 1$.*

Proof of Lemma A.5: When $\beta \rightarrow 1$, $\mu(1, 0) \rightarrow \mu(1)$ and $\mu(0, 0) \rightarrow \mu(0)$, defined in Lemmas A.4 and A.1 respectively. The politician's cutoff $\tilde{x}^*$ solves

$$2\eta(\tilde{x}^*) - 1 + \Delta(0; \tilde{x}^*) = 0,$$

where $\Delta(0) = F(\mu(1, 0)) - F(\mu(0, 0)) \rightarrow F(\mu(1)) - F(\mu(0))$. ■

Lemma A.6 *The probability of climate action is inverse U-shaped in β .*

Proof of Lemma A.6: Define $I(\tilde{x})$ as

$$I(\tilde{x}) = 2\eta(\tilde{x}) - 1 + \eta(\tilde{x})(1 - \beta)\Delta(1; \tilde{x}) + (1 - \eta(\tilde{x}) + \eta(\tilde{x})\beta)\Delta(0; \tilde{x}).$$

By the implicit function theorem,

$$\frac{d\tilde{x}^*}{d\beta} = -\frac{\partial I(\tilde{x})/\partial\beta}{\partial I(\tilde{x})/\partial\tilde{x}}.$$

By definition of the equilibrium cutoff, $I(\tilde{x})$ is increasing in $\tilde{x}$. Partially differentiating with respect to β gives

$$\frac{\partial I(\tilde{x})}{\partial\beta} = -\eta(\tilde{x})(\Delta(1; \tilde{x}) - \Delta(0; \tilde{x})) + (1 - \eta(\tilde{x}) + \eta(\tilde{x})\beta)\frac{\partial\Delta(0; \tilde{x})}{\partial\beta}.$$

The first term is negative and the second term is positive because $\mu(1, 0; \tilde{x})$ is increasing in β and $\mu(0, 0; \tilde{x})$ is decreasing in β . Note that at $\beta = 0$, $\mu(1, 0; \tilde{x}) = 0$ so $\Delta(0; \tilde{x})$ is at its minimum. But $\frac{\partial\Delta(0; \tilde{x})}{\partial\beta}$ is at a maximum because $\frac{\partial\Delta(0; \tilde{x})}{\partial\beta}$ is decreasing in β :

$$\begin{aligned} \frac{\partial^2\Delta(0; \tilde{x})}{\partial\beta^2} &= \frac{\partial^2\mu(1, 0; \tilde{x})}{\partial\beta^2}F'(\mu(1, 0; \tilde{x})) + \left(\frac{\partial\mu(1, 0; \tilde{x})}{\partial\beta}\right)^2F''(\mu(1, 0; \tilde{x})) \\ &\quad - \frac{\partial^2\mu(0, 0; \tilde{x})}{\partial\beta^2}F'(\mu(0, 0; \tilde{x})) - \left(\frac{\partial\mu(0, 0; \tilde{x})}{\partial\beta}\right)^2F''(\mu(0, 0; \tilde{x})) < 0, \end{aligned}$$

because $F(\cdot)$ has a log-concave density and $\frac{\partial^2\mu(1, 0; \tilde{x})}{\partial\beta^2} < 0$ and $\frac{\partial^2\mu(0, 0; \tilde{x})}{\partial\beta^2} > 0$. Since $\frac{\partial\Delta(0; \tilde{x})}{\partial\beta}$ is maximized at $\beta = 0$ and is scaled by $(1 - \eta(\tilde{x})) > 0$, the second term is strictly positive and dominates the first term. Hence $\frac{\partial I}{\partial\beta} > 0$ at $\beta = 0$.

Furthermore, as $\beta \rightarrow 1$, the posterior beliefs converge to those in the model without the interest group from Lemma A.5. This means that $\frac{\partial\Delta(0; \tilde{x})}{\partial\beta} \rightarrow 0$ and $\frac{\partial I(\tilde{x})}{\partial\beta} < 0$ for large β . Thus, $\frac{d\tilde{x}^*}{d\beta}$ is U-shaped in β by the implicit function theorem.

The probability of climate action is

$$A(\tilde{x}^*) = \pi(1 - G(\tilde{x}^*; 1)) + (1 - \pi)(1 - G(\tilde{x}^*; 0)).$$

Differentiating with respect to β gives

$$\frac{dA(\tilde{x}^*)}{d\beta} = -\pi g(\tilde{x}^*; 1) \frac{d\tilde{x}^*}{d\beta} - (1 - \pi) g(\tilde{x}^*; 0) \frac{d\tilde{x}^*}{d\beta}.$$

As $\frac{d\tilde{x}^*}{d\beta}$ is U-shaped in β , $\frac{dA(\tilde{x}^*)}{d\beta}$ is inverse U-shaped in β . ■

Lemma A.7 *There exists a unique $\pi^\dagger$ such that the special interest's optimal misreporting strategy is*

$$\beta^*(\pi) = \begin{cases} 1 & \pi < \pi^\dagger \\ 0 & \pi > \pi^\dagger. \end{cases}$$

Proof of Lemma A.7: The special interest has the objective function

$$\max_{\beta \in [0,1]} 1 - A(\tilde{x}^*).$$

By Lemma A.6, $\tilde{x}^*$ is U-shaped in β so $1 - A(\tilde{x}^*)$ is also U-shaped in β . This implies that the maximum lies either at $\beta = 0$ or $\beta = 1$.

Let $\tilde{x}_\beta^*$ be the politician's optimal cutoff for a misreporting level β . Observing what happens at each of the endpoints, we see that at $\beta = 0$, $\tilde{x}_0^*$ solves

$$\begin{aligned} 2\eta(\tilde{x}_0^*) - 1 + \eta(\tilde{x}_0^*)\Delta(1) + (1 - \eta(\tilde{x}_0^*))\Delta(0) &= 0 \\ (2\eta(\tilde{x}_0^*) - 1)(1 + \Delta(1)) &= 0 \\ \eta(\tilde{x}_0^*) &= \frac{1}{2}. \end{aligned}$$

The first line comes from direct substitution for $\beta = 0$, the second line uses the fact that $\Delta(0) = F(0) - F(1) = -\Delta(1)$ when $\beta = 0$, and the third line rearranges.

When $\beta = 1$, the politician's problem is the same as the model without the interest group from Lemma A.5,

$$\begin{aligned} 2\eta(\tilde{x}_1^*) - 1 + \underbrace{F(\mu(1)) - F(\mu(0))}_{\Delta} &= 0 \\ \eta(\tilde{x}_1^*) &= \frac{1 - \Delta}{2}. \end{aligned}$$

To order the cutpoints, we simply need to know the sign of Δ . By Lemma A.3, we know that $\mu(1) > \mu(0)$ and thus $\Delta > 0$ iff $\pi > \pi^\dagger$, so when $\pi > \pi^\dagger$ we have $\tilde{x}_1^* < \tilde{x}_0^*$. Symmetrically, $\mu(1) < \mu(0)$ and $\Delta < 0$ iff $\pi < \pi^\dagger$, yielding $\tilde{x}_0^* < \tilde{x}_1^*$. Hence when $\pi < \pi^\dagger$ the special interest group sends $\beta = 1$ and when $\pi > \pi^\dagger$ the special interest group sends $\beta = 0$. ■ ■

Effects of Foreign Climate Action

Proof of Proposition 3: I proceed in several parts. Lemma A.8 show the existence of a unique equilibrium, holding β fixed. Lemmas A.9, A.10, and A.11 establish international analogs of Lemmas A.2 and A.6. The first bullet of the proposition is proven in Lemma A.12. The second bullet of the proposition is proven in Lemma A.13, and the third bullet in Lemma A.14.

Lemma A.8 *Let $\mathcal{I} = \{a_i, s, a_j\}$. There exists a unique, symmetric cutoff $\tilde{x}^*$ that characterizes the equilibrium.*

Proof of Lemma A.8: The voter in country i observes the triple (a_i, s, a_j) and forms posterior belief $\mu_i(a_i, s, a_j) = P(\omega = a_i | a_i, s, a_j)$. The voter reelects politician i iff $\mu_i(a_i, s, a_j) \geq \varepsilon_i$, or with probability $F(\mu_i(a_i, s, a_j))$.

Let $p_{j\omega} = P(a_j = 1 | \omega)$ be the belief that politician j chooses $a_j = 1$ in state ω for actors in country i . Write $\Delta_i(s, a_j) = F(\mu_i(1, s, a_j)) - F(\mu_i(0, s, a_j))$. Then the politician plays $a_i = 1$ iff

$$\begin{aligned} \eta(x_i) & \left[1 + p_{j1}\beta\Delta_i(0, 1) + p_{j1}(1 - \beta)\Delta_i(1, 1) + (1 - p_{j1})(1 - \beta)\Delta_i(1, 0) + (1 - p_{j1})\beta\Delta_i(0, 0) \right] \\ & \geq (1 - \eta(x_i)) \left[1 - p_{j0}\Delta_i(0, 1) - (1 - p_{j0})\Delta_i(0, 0) \right], \end{aligned}$$

which can be rearranged to

$$\frac{g(x_i; 1)}{g(x_i; 0)} \geq \frac{1 - \pi}{\pi} \frac{1 - p_{j0}\Delta_i(0, 1) - (1 - p_{j0})\Delta_i(0, 0)}{1 + p_{j1}\beta\Delta_i(0, 1) + p_{j1}(1 - \beta)\Delta_i(1, 1) + (1 - p_{j1})(1 - \beta)\Delta_i(1, 0) + (1 - p_{j1})\beta\Delta_i(0, 0)}.$$

Now the RHS is constant in x_i and the LHS is increasing in x_i such that there is a solution $\tilde{x}_i$ satisfying with equality. Analogously for j , there is a $\tilde{x}_j$. This means that politician i 's best response to politician j and voter i is a cutoff rule. Then the politician plays $a = 1$ when $x \geq \tilde{x}$ for each respective country. Moreover, country i 's beliefs about the behavior of country j are $p_{j\omega} = 1 - G(\tilde{x}_j; \omega)$, which is decreasing in $\tilde{x}_j$.

Given cutoffs $(\tilde{x}_i, \tilde{x}_j)$, the voter has posterior beliefs $\mu_i(a_i, s, a_j; \tilde{x}_i, \tilde{x}_j)$:

$$\begin{aligned} \mu_i(1, 1, a_j; \tilde{x}_i, \tilde{x}_j) &= 1. \\ \mu_i(0, 1, a_j; \tilde{x}_i, \tilde{x}_j) &= 0. \\ \mu_i(1, 0, 1; \tilde{x}_i, \tilde{x}_j) &= \frac{\pi\beta p_{j1}(1 - G(\tilde{x}_i; 1))}{\pi\beta p_{j1}(1 - G(\tilde{x}_i; 1)) + (1 - \pi)p_{j0}(1 - G(\tilde{x}_i; 0))}. \\ \mu_i(1, 0, 0; \tilde{x}_i, \tilde{x}_j) &= \frac{\pi\beta(1 - p_{j1})(1 - G(\tilde{x}_i; 1))}{\pi\beta(1 - p_{j1})(1 - G(\tilde{x}_i; 1)) + (1 - \pi)(1 - p_{j0})(1 - G(\tilde{x}_i; 0))}. \\ \mu_i(0, 0, 1; \tilde{x}_i, \tilde{x}_j) &= \frac{(1 - \pi)p_{j0}G(\tilde{x}_i; 0)}{(1 - \pi)p_{j0}G(\tilde{x}_i; 0) + \pi\beta p_{j1}G(\tilde{x}_i; 1)}. \\ \mu_i(0, 0, 0; \tilde{x}_i, \tilde{x}_j) &= \frac{(1 - \pi)(1 - p_{j0})G(\tilde{x}_i; 0)}{(1 - \pi)(1 - p_{j0})G(\tilde{x}_i; 0) + \pi\beta(1 - p_{j1})G(\tilde{x}_i; 1)}. \end{aligned}$$

Since $\mu_i(1, 1, 1; \tilde{x}_i, \tilde{x}_j) = \mu_i(1, 1, 0; \tilde{x}_i, \tilde{x}_j)$ and $\mu_i(0, 1, 1; \tilde{x}_i, \tilde{x}_j) = \mu_i(0, 1, 0; \tilde{x}_i, \tilde{x}_j)$, it follows that $\Delta_i(1, 1) = \Delta_i(1, 0) = \Delta_i(1)$. It is evident that $\mu_i(1, s, a_j; \tilde{x}_i, \tilde{x}_j)$ is increasing in $\tilde{x}_i$ and $\mu_i(0, s, a_j; \tilde{x}_i, \tilde{x}_j)$ is decreasing in $\tilde{x}_i$ so $\Delta_i(s_i, a_j)$ is increasing in $\tilde{x}_i$. Then the politician's cutoff $\tilde{x}_i$, given any $\tilde{x}_j$, solves

$$2\eta(\tilde{x}_i) - 1 + (1 - \beta)\eta(\tilde{x}_i)\Delta_i(1) + (\beta\eta(\tilde{x}_i)p_{j1} + (1 - \eta(\tilde{x}_i))p_{j0})\Delta_i(0, 1; \tilde{x}_i, \tilde{x}_j) + (\beta\eta(\tilde{x}_i)(1 - p_{j1}) + (1 - \eta(\tilde{x}_i))(1 - p_{j0}))\Delta_i(0, 0; \tilde{x}_i, \tilde{x}_j) = 0. \quad (3)$$

Define $I(\tilde{x}_i, \tilde{x}_j)$ as the politician's constraint as in Equation 3. Now imposing symmetry such that $\tilde{x}_i = \tilde{x}_j = \tilde{x}$, the equilibrium cutoff solves

$$\begin{aligned} I(\tilde{x}, \tilde{x}) = & 2\eta(\tilde{x}) - 1 + \eta(\tilde{x})(1 - \beta)\Delta(1) \\ & + \left(\eta(\tilde{x})p_{j1}(\tilde{x})\beta + (1 - \eta(\tilde{x}))p_{j0}(\tilde{x}) \right) \Delta(0, 1; \tilde{x}, \tilde{x}) \\ & + \left(\eta(\tilde{x})(1 - p_{j1}(\tilde{x}))\beta + (1 - \eta(\tilde{x}))(1 - p_{j0}(\tilde{x})) \right) \Delta(0, 0; \tilde{x}, \tilde{x}). \end{aligned}$$

Differentiating with respect to $\tilde{x}$ (and suppressing dependence on $\tilde{x}$ to economize notation) gives

$$\begin{aligned} \frac{\partial I(\tilde{x}, \tilde{x})}{\partial \tilde{x}} = & \left(2 + (1 - \beta)\Delta(1) \right) \frac{\partial \eta}{\partial \tilde{x}} + \left(\frac{\partial \eta}{\partial \tilde{x}}(\beta p_{j1} - p_{j0}) + \eta\beta \frac{\partial p_{j1}}{\partial \tilde{x}} + (1 - \eta) \frac{\partial p_{j0}}{\partial \tilde{x}} \right) \Delta(0, 1) \\ & + \left(\eta\beta p_{j1} + (1 - \eta)p_{j0} \right) \frac{\partial \Delta(0, 1)}{\partial \tilde{x}} + \left(\eta\beta(1 - p_{j1}) + (1 - \eta)(1 - p_{j0}) \right) \frac{\partial \Delta(0, 0)}{\partial \tilde{x}} \\ & + \left(\frac{\partial \eta}{\partial \tilde{x}}(\beta(1 - p_{j1}) - (1 - p_{j0})) - \eta\beta \frac{\partial p_{j1}}{\partial \tilde{x}} - (1 - \eta) \frac{\partial p_{j0}}{\partial \tilde{x}} \right) \Delta(0, 0). \end{aligned}$$

Since MLRP implies hazard rate ordering, it can be shown that $\mu_i(1, 0, a_j; \tilde{x}, \tilde{x})$ is increasing in $\tilde{x}$ and $\mu_i(0, 0, a_j; \tilde{x}, \tilde{x})$ is decreasing in $\tilde{x}$. Hence $\Delta(0, a_j)$ is increasing in $\tilde{x}$. Regrouping terms,

$$\begin{aligned} \frac{\partial I(\tilde{x}, \tilde{x})}{\partial \tilde{x}} = & \left(2 + (1 - \beta)\Delta(1) + (\beta p_{j1} - p_{j0})\Delta(0, 1) + (\beta(1 - p_{j1}) - (1 - p_{j0}))\Delta(0, 0) \right) \frac{\partial \eta}{\partial \tilde{x}} \\ & + \left(\eta\beta p_{j1} + (1 - \eta)p_{j0} \right) \frac{\partial \Delta(0, 1)}{\partial \tilde{x}} + \left(\eta\beta(1 - p_{j1}) + (1 - \eta)(1 - p_{j0}) \right) \frac{\partial \Delta(0, 0)}{\partial \tilde{x}} \\ & + \left(\eta\beta \frac{\partial p_{j1}}{\partial \tilde{x}} + (1 - \eta) \frac{\partial p_{j0}}{\partial \tilde{x}} \right) (\Delta(0, 1) - \Delta(0, 0)). \end{aligned}$$

The first line is positive because in equilibrium $\Delta(1) \geq \Delta(0, 1) \geq \Delta(0, 0)$ and each Δ is bounded in $[-1, 1]$. The second line is also positive. The third line is negative because $\Delta(0, 1) \geq \Delta(0, 0)$. It is thus sufficient to demonstrate that the positive, informational direct effect of increasing $\tilde{x}$ given by the first line exceeds the negative, strategic effect given in the third line.

Observe that $\frac{\partial \eta}{\partial \tilde{x}}$ can be decomposed as

$$\frac{\partial \eta}{\partial \tilde{x}} = \eta(1 - \eta) \left(\frac{g'(\tilde{x}; 1)}{g(\tilde{x}; 1)} - \frac{g'(\tilde{x}; 0)}{g(\tilde{x}; 0)} \right) > 0,$$

where $g'(x; \omega)$ is the derivative of the density $g(x; \omega)$. Let $C = 2 + (1 - \beta)\Delta(1) + (\beta p_{j1} - p_{j0})\Delta(0, 1) + (\beta(1 - p_{j1}) - (1 - p_{j0}))\Delta(0, 0)$. Then the informational effect dominates the strategic effect iff

$$\begin{aligned} \eta(1 - \eta) \left(\frac{g'(\tilde{x}; 1)}{g(\tilde{x}; 1)} - \frac{g'(\tilde{x}; 0)}{g(\tilde{x}; 0)} \right) C &> \left(\eta \beta g(\tilde{x}; 1) + (1 - \eta) g(\tilde{x}; 0) \right) \left(\Delta(0, 1) - \Delta(0, 0) \right) \\ \Leftrightarrow \left(\frac{g'(\tilde{x}; 1)}{g(\tilde{x}; 1)} - \frac{g'(\tilde{x}; 0)}{g(\tilde{x}; 0)} \right) &> \left(\frac{\beta g(\tilde{x}; 1)}{1 - \eta} + \frac{g(\tilde{x}; 0)}{\eta} \right) \frac{\Delta(0, 1) - \Delta(0, 0)}{C}. \end{aligned}$$

The LHS represents the speed with which the politician's beliefs about ω change. The RHS is the strategic uncertainty of the foreign politician's action. I assume that this holds in order to ensure a unique equilibrium; if the signal distribution $G(\cdot; \omega)$ is the normal distribution with mean ω and variance σ^2 , the LHS is simply $\frac{1}{\sigma^2}$ and there is an upper bound $\sigma < \bar{\sigma}$ that can be parameterized. Thus, since $\frac{\partial I(\tilde{x}, \tilde{x})}{\partial \tilde{x}} > 0$, there exists a unique $\tilde{x}^*$ such that each politician chooses $a_i = 1$ iff $x_i \geq \tilde{x}^*$. ■

Lemma A.9 *The threshold $\tilde{x}^*$ is decreasing in π .*

Proof of Lemma A.9: By the implicit function theorem,

$$\frac{d\tilde{x}^*}{d\pi} = - \frac{\partial I(\tilde{x}, \tilde{x}) / \partial \pi}{\partial I(\tilde{x}, \tilde{x}) / \partial \tilde{x}}.$$

Partially differentiating with respect to π gives

$$\begin{aligned} \frac{\partial I(\tilde{x}, \tilde{x})}{\partial \pi} &= \frac{\partial \eta}{\partial \pi} \left[2 + (1 - \beta)\Delta(1) + (\beta p_{j1} - p_{j0})\Delta(0, 1) + (\beta(1 - p_{j1}) - (1 - p_{j0}))\Delta(0, 0) \right] \\ &\quad + (\beta \eta p_{j1} + (1 - \eta) p_{j0}) \frac{\partial \Delta(0, 1)}{\partial \pi} + (\beta \eta (1 - p_{j1}) + (1 - \eta)(1 - p_{j0})) \frac{\partial \Delta(0, 0)}{\partial \pi}. \end{aligned}$$

The first line is positive because $\frac{\partial \eta}{\partial \pi} > 0$ by MLRP and the term in brackets is positive akin to the argument in Lemma A.8. Since $\mu_i(1, 0, a_j; \tilde{x}, \tilde{x})$ is increasing in π and $\mu_i(0, 0, a_j; \tilde{x}, \tilde{x})$ is decreasing in π , we have $\frac{\partial \Delta(0, a_j)}{\partial \pi} \geq 0$ so the third line is positive as well. Then $\frac{\partial I(\tilde{x}, \tilde{x})}{\partial \pi} \geq 0$ and by the implicit function theorem $\frac{d\tilde{x}^*}{d\pi} \leq 0$. ■

Lemma A.10 *The threshold $\tilde{x}^*$ is U-shaped in β .*

Proof of Lemma A.10: This proof is analogous to Lemma A.6. By the implicit function theorem,

$$\frac{d\tilde{x}^*}{d\beta} = - \frac{\partial I(\tilde{x}, \tilde{x}) / \partial \beta}{\partial I(\tilde{x}, \tilde{x}) / \partial \tilde{x}}.$$

The denominator is positive at the equilibrium threshold $\tilde{x}^*$ as shown in Lemma A.8. Partially differentiating with respect to β yields

$$\begin{aligned} \frac{\partial I(\tilde{x}, \tilde{x})}{\partial \beta} = & -\eta \left(\Delta(1) - p_{j1} \Delta(0, 1) - (1 - p_{j1}) \Delta(0, 0) \right) \\ & + \left(\eta p_{j1} \beta + (1 - \eta) p_{j0} \right) \frac{\partial \Delta(0, 1)}{\partial \beta} + \left(\eta (1 - p_{j1}) \beta + (1 - \eta) (1 - p_{j0}) \right) \frac{\partial \Delta(0, 0)}{\partial \beta}. \end{aligned}$$

It can be shown that $\mu_i(1, 0, a_j)$ are increasing in β and $\mu_i(0, 0, a_j)$ are decreasing in β . Then $\Delta(0, a_j)$ is also increasing in β_i . By definition of posterior beliefs, $\Delta_i(1) \geq \Delta(0, 1) \geq \Delta(0, 0)$. This means that first term is negative and the second and third terms are positive. Observe that this expression is exactly the same as in Lemma A.6 except it also incorporates the behavior of politician j . Then, as in Lemma A.6, it can be shown that $I(\tilde{x}, \tilde{x})$ is increasing in β for small β and decreasing in β for large β . Then by the implicit function theorem $\frac{d\tilde{x}^*}{d\beta}$ is U-shaped. ■

Lemma A.11 *There exists a unique $\pi_{intl}^\dagger$ such that the special interest's optimal misreporting strategy is*

$$\beta^*(\pi) = \begin{cases} 1 & \pi < \pi_{intl}^\dagger \\ 0 & \pi > \pi_{intl}^\dagger. \end{cases}$$

Proof of Lemma A.11: The special interest group has the objective function

$$\max_{\beta \in [0, 1]} 2 - A_i(\tilde{x}^*) - A_j(\tilde{x}^*),$$

where $A_i(\tilde{x}^*) = P(a_i = 1 | \tilde{x}^*) = \pi(1 - G(\tilde{x}^*; 1)) + (1 - \pi)(1 - G(\tilde{x}^*; 0))$. By symmetry, this reduces to

$$\max_{\beta \in [0, 1]} \pi G(\tilde{x}^*; 1) + (1 - \pi) G(\tilde{x}^*; 0).$$

Analogous to Lemma A.7, since $\tilde{x}^*$ is U-shaped in β , the special interest group's choice of β must either be $\beta^* = 0$ or $\beta^* = 1$. Let $\tilde{x}_\beta^*$ be the optimal threshold at a given β .

If $\beta^* = 0$, the politician's threshold $\tilde{x}^*$ solves

$$2\eta(\tilde{x}_0^*) - 1 + \eta(\tilde{x}_0^*)\Delta(1) + (1 - \eta(\tilde{x}_0^*))p_{j0}\Delta(0, 1) + (1 - \eta(\tilde{x}_0^*))(1 - p_{j0})\Delta(0, 0) = 0.$$

Notice that when $\beta_i = 0$, $\Delta(0, 1) = \Delta(0, 0) = F(0) - F(1) = -\Delta(1)$. Simplifying yields that $\eta(\tilde{x}_0^*) = \frac{1}{2}$.

Similarly, if $\beta = 1$, the cutoff $\tilde{x}_1^*$ solves

$$2\eta(\tilde{x}_1^*) - 1 + (\eta(\tilde{x}_1^*)p_{j1} + (1 - \eta(\tilde{x}_1^*))p_{j0})\Delta(0, 1) + (\eta(\tilde{x}_1^*)(1 - p_{j1}) + (1 - \eta(\tilde{x}_1^*))(1 - p_{j0}))\Delta(0, 0) = 0.$$

Call the LHS of this equation $\mathcal{B}(\tilde{x})$, which is the politician's indifference condition when $\beta = 1$. From Lemma A.8 it is clear that $\mathcal{B}(\tilde{x})$ is increasing in $\tilde{x}$, where the zero is at $\tilde{x}_1^*$. If

we were to evaluate $\mathcal{B}(\tilde{x})$ at any other $\tilde{x}$, we would know whether such a $\tilde{x}$ is greater than or less than $\tilde{x}_1^*$ based on the sign of $\mathcal{B}(\tilde{x})$.

Evaluating $\mathcal{B}(\tilde{x}_0^*)$ and noting from above that $\eta(\tilde{x}_0^*) = \frac{1}{2}$, we have

$$\mathcal{B}(\tilde{x}_0^*) = \bar{p}_j \Delta(0, 1) + (1 - \bar{p}_j) \Delta(0, 0),$$

where $\bar{p}_j = \frac{p_{j1} + p_{j0}}{2}$. If $\mathcal{B}(\tilde{x}_0^*) < 0$, then $\tilde{x}_0^* < \tilde{x}_1^*$ and the special interest group chooses $\beta^* = 1$. If $\mathcal{B}(\tilde{x}_0^*) > 0$, then $\tilde{x}_0^* > \tilde{x}_1^*$ and the special interest group chooses $\beta^* = 0$. Differentiating with respect to π gives

$$\frac{d\mathcal{B}(\tilde{x}_0^*)}{d\pi} = \frac{d\bar{p}_j}{d\pi} (\Delta(0, 1) - \Delta(0, 0)) + \bar{p}_j \frac{d\Delta(0, 1)}{d\pi} + (1 - \bar{p}_j) \frac{d\Delta(0, 0)}{d\pi} > 0.$$

At $\pi = 0$, $\mathcal{B}(\tilde{x}_0^*) = \Delta(0, 0)|_{\pi=0} = F(0) - F(1) < 0$. At $\pi = 1$, $\mathcal{B}(\tilde{x}_1^*) = \Delta(0, 1)|_{\pi=1} = F(1) - F(0) > 0$. Hence there is a unique $\pi_{intl}^\dagger$ such that $\mathcal{B}(\tilde{x}_0^*) = 0$ and where $\tilde{x}_0^* = \tilde{x}_1^*$. When $\pi < \pi_{intl}^\dagger$, $\tilde{x}_0^* < \tilde{x}_1^*$ and $\beta^* = 1$. When $\pi > \pi_{intl}^\dagger$, $\tilde{x}_0^* > \tilde{x}_1^*$ and $\beta^* = 0$.

By state-symmetry of $g(\cdot; \omega)$, we can further characterize $\pi_{intl}^\dagger = \frac{1}{2}$. The point of indifference, i.e. at $\pi = \pi_{intl}^\dagger$, is where $\mathcal{B}(\tilde{x}_0^*) = \bar{p}_j \Delta(0, 1) + (1 - \bar{p}_j) \Delta(0, 0) = 0$. Suppose that $\pi_{intl}^\dagger = \frac{1}{2}$ and $\tilde{x}_0^* = c$. By state-symmetry, $G(c; 1) + G(c; 0) = 1$. Then $p_{j1} = 1 - G(c; 1) = G(c; 0)$ and $p_{j0} = 1 - G(c; 0) = G(c; 1)$, and $\bar{p}_j = \frac{1}{2}$. We now have that at $\pi = \pi_{intl}^\dagger$,

$$\Delta(0, 1) + \Delta(0, 0) = 0.$$

To verify this, observe that at $\pi = \frac{1}{2}$, $\beta = 1$, and $\tilde{x} = c$, the posterior beliefs $\mu_i(1, 0, a_j; c, c)$ and $\mu_i(0, 0, a_j; c, c)$ satisfy

$$\begin{aligned} \mu_i(1, 0, 1; c, c) &= \mu_i(0, 0, 0; c, c) = \frac{G(c; 0)^2}{G(c; 0)^2 + G(c; 1)^2}. \\ \mu_i(1, 0, 0; c, c) &= \mu_i(0, 0, 1; c, c) = \frac{1}{2}. \end{aligned}$$

Then $\Delta(0, 1) = F\left(\frac{G(c; 0)^2}{G(c; 0)^2 + G(c; 1)^2}\right) - F\left(\frac{1}{2}\right)$ and $\Delta(0, 0) = F\left(\frac{1}{2}\right) - F\left(\frac{G(c; 0)^2}{G(c; 0)^2 + G(c; 1)^2}\right)$, as desired. Hence since $\pi_{intl}^\dagger$ is unique, $\mathcal{B}(\tilde{x}_0^*) = 0$ iff $\pi = \pi_{intl}^\dagger = \frac{1}{2}$. ■

Lemma A.12 *Countries' thresholds can either be strategic complements or strategic substitutes.*

Proof of Lemma A.12: To study strategic complementarity and substitutability, consider the effect of changing $\tilde{x}_j$ prior to imposing symmetry of cutoffs. By the implicit function theorem, strategic effects are given by

$$\frac{d\tilde{x}_i}{d\tilde{x}_j} = -\frac{\partial I(\tilde{x}_i, \tilde{x}_j)/\partial \tilde{x}_j}{\partial I(\tilde{x}_i, \tilde{x}_j)/\partial \tilde{x}_i},$$

where the denominator is positive by Lemma A.8. Partially differentiating with respect to $\tilde{x}_j$ gives

$$\begin{aligned} \frac{\partial I(\tilde{x}_i, \tilde{x}_j)}{\partial \tilde{x}_j} &= \left(\beta \eta(\tilde{x}_i) \frac{\partial p_{j1}}{\partial \tilde{x}_j} + (1 - \eta(\tilde{x}_i)) \frac{\partial p_{j0}}{\partial \tilde{x}_j} \right) (\Delta_i(0, 1) - \Delta_i(0, 0)) \\ &\quad + (\beta \eta(\tilde{x}_i) p_{j1} + (1 - \eta(\tilde{x}_i)) p_{j0}) \frac{\partial \Delta_i(0, 1)}{\partial \tilde{x}_j} + (\beta \eta(\tilde{x}_i)(1 - p_{j1}) + (1 - \eta(\tilde{x}_i))(1 - p_{j0})) \frac{\partial \Delta_i(0, 0)}{\partial \tilde{x}_j}. \end{aligned}$$

The first line is negative because $\frac{\partial p_{j\omega}}{\partial \tilde{x}_j} < 0$ and $\Delta_i(0, 1) \geq \Delta_i(0, 0)$. The second line is positive as the MLRP implies the hazard rate ordering: $\mu_i(1, 0, a_j; \tilde{x}_i, \tilde{x}_j)$ is increasing in $\tilde{x}_j$, $\mu_i(0, 0, a_j; \tilde{x}_i, \tilde{x}_j)$ is decreasing in $\tilde{x}_j$, so $\Delta_i(0, a_j)$ is increasing in $\tilde{x}_j$. This cannot be signed in general, but when the first term dominates, $\frac{\partial I(\tilde{x}_i, \tilde{x}_j)}{\partial \tilde{x}_j} < 0$ and actions are strategic complements, $\frac{d\tilde{x}_i}{d\tilde{x}_j} \geq 0$. If the second and third terms dominate, $\frac{\partial I(\tilde{x}_i, \tilde{x}_j)}{\partial \tilde{x}_j} > 0$ and actions are strategic substitutes, $\frac{d\tilde{x}_i}{d\tilde{x}_j} \leq 0$. ■

Lemma A.13 *Call $\tilde{x}_D^*$ the optimal threshold that the politician sets in the domestic game, characterized in Lemma A.4, and call $\tilde{x}_I^*$ the optimal threshold that the politician sets in the international game, characterized in Lemma A.8. When $\pi < \pi_{intl}^\dagger$, $\tilde{x}_I^* < \tilde{x}_D^*$.*

Proof of Lemma A.13: Observe that at $\pi = 0$, $\tilde{x}_D^* = \tilde{x}_I^* \rightarrow \infty$. Moreover, by state-symmetry, at $\pi^\dagger = \pi_{intl}^\dagger = \frac{1}{2}$, $\tilde{x}_D^* = \tilde{x}_I^* = c$. On the interior, define $\mathcal{B}_D(\tilde{x}_D^*) = 0$ such that $\tilde{x}_D^*$ solves

$$2\eta(\tilde{x}_D^*) - 1 + \Delta(0; \tilde{x}_D^*) = 0,$$

and define $\mathcal{B}_I(\tilde{x}_I^*) = 0$ $\tilde{x}_I^*$ solves

$$\begin{aligned} 2\eta(\tilde{x}_I^*) - 1 + (\eta(\tilde{x}_I^*) p_{j1}(\tilde{x}_I^*) + (1 - \eta(\tilde{x}_I^*) p_{j0}(\tilde{x}_I^*)) \Delta(0, 1; \tilde{x}_I^*, \tilde{x}_I^*) \\ + (\eta(\tilde{x}_I^*)(1 - p_{j1}(\tilde{x}_I^*)) + (1 - \eta(\tilde{x}_I^*)(1 - p_{j0}(\tilde{x}_I^*)) \Delta(0, 0; \tilde{x}_I^*, \tilde{x}_I^*) = 0. \end{aligned}$$

Observe that $\mathcal{B}_I(\cdot)$ is identical to $\mathcal{B}(\tilde{x}_0^*)$ in Lemma A.11, and the proof follows an analogous structure. Since $\mathcal{B}_I(\cdot)$ is increasing in its argument by Lemma A.8, if $\mathcal{B}_I(\tilde{x}_D^*) > 0$, then $\tilde{x}_I^* < \tilde{x}_D^*$. Observe that $\tilde{x}_D^*$, we have $\eta(\tilde{x}_D^*) = \frac{1 - \Delta(0)}{2} > 0$ and when $\pi < \pi^\dagger$, $\Delta(0) < 0$. Substituting in and simplifying yields

$$\mathcal{B}_I(\tilde{x}_D^*) = -\Delta(0) + \frac{\Delta(0, 1) - \Delta(0, 0)}{2} (p_{j1} + p_{j0} - \Delta(0)(p_{j1} - p_{j0})) + \Delta(0, 0).$$

It suffices to show that $-\Delta(0) > \Delta(0, 0)$ at $\pi < \pi^\dagger$. Then we need $F(\mu(0, 0)) - F(\mu(1, 0)) \geq F(\mu(1, 0, 0)) - F(\mu(0, 0, 0))$. By definition of posterior beliefs, $\mu(0, 0) \geq \mu(1, 0, 0)$ and $\mu(0, 0, 0) \geq \mu(1, 0)$ iff $\frac{(1-\pi)^2}{\pi^2} \geq \frac{(1-p_{j1})G(\tilde{x}_D^*; 1)(1-G(\tilde{x}_D^*; 1))}{(1-p_{j0})G(\tilde{x}_D^*; 0)(1-G(\tilde{x}_D^*; 0))}$, which is true for $\pi < \pi^\dagger$. Hence for $\pi < \pi^\dagger$, $\mathcal{B}_I(\tilde{x}_D^*) > 0$ and $\tilde{x}_I^* < \tilde{x}_D^*$, as desired. ■

Lemma A.14 *Let $A(\tilde{x}^*)$ be the probability of climate action. Then define bilateral climate action as $A(\tilde{x}^*)^2$, bilateral climate inaction as $(1 - A(\tilde{x}^*))^2$ and unilateral climate action*

as $2A(\tilde{x}^*)(1 - A(\tilde{x}^*))$. Then bilateral climate action is increasing in π , bilateral climate inaction is decreasing in π , and unilateral climate action is increasing in π when $\pi < \pi^\dagger$ and decreasing in π when $\pi > \pi^\dagger$.

Proof of Lemma A.14: From Lemma A.9, $\tilde{x}^*$ is decreasing in π and so the probability of climate action is increasing in π (see also Lemma A.2).

Differentiating with respect to π , bilateral climate action is increasing: $2A(\tilde{x}^*)\frac{dA(\tilde{x}^*)}{d\pi} \geq 0$ and bilateral climate inaction is decreasing: $-2(1 - A(\tilde{x}^*))\frac{dA(\tilde{x}^*)}{d\pi} \leq 0$. The derivative of unilateral climate action with respect to π is $2(1 - A(\tilde{x}^*))\frac{dA(\tilde{x}^*)}{d\pi}$, where the sign depends on whether $A(\tilde{x}^*)$ is above or below $\frac{1}{2}$. Observe that when $\pi = \pi^\dagger = \frac{1}{2}$, we have that $A(\tilde{x}^*)$ such that for any $\pi < \pi^\dagger$, $A(\tilde{x}^*) < \frac{1}{2}$ and the expression is increasing, while for any $\pi > \pi^\dagger$, $A(\tilde{x}^*) > \frac{1}{2}$ and the expression is decreasing:

$$\begin{aligned} A(\tilde{x}^*)|_{\pi=\pi^\dagger} &= \frac{1}{2}(1 - G(\tilde{x}^*; 1)) + \frac{1}{2}(1 - G(\tilde{x}^*; 0)) \\ &= \frac{1}{2}G(c; 0) + \frac{1}{2}G(c; 1) \\ &= \frac{1}{2}. \end{aligned}$$

■ ■

Remark A.1 provides the theoretical underpinnings for the comparisons of beliefs examined in Figure 6. These relationships are immediate from Lemma A.8, but considering that they are the formal scaffolding for the empirical analysis it is useful to list them exhaustively.

Remark A.1 *The following comparisons hold in equilibrium:*

- *Domestic comparisons* ($\mu_i(1, s, a_j) - \mu_i(0, s, a_j)$):

1. $\mu_i(1, 1, 1) - \mu_i(0, 1, 1) > 0$;
2. $\mu_i(1, 1, 0) - \mu_i(0, 1, 0) > 0$;
3. $\mu_i(1, 0, 1) - \mu_i(0, 0, 1)$ is ambiguously signed;
4. $\mu_i(1, 0, 0) - \mu_i(0, 0, 0) < 0$.

- *Signal comparisons* ($\mu_i(a_i, 1, a_j) - \mu_i(a_i, 0, a_j)$):

1. $\mu_i(1, 1, 1) - \mu_i(1, 0, 1) > 0$;
2. $\mu_i(1, 1, 0) - \mu_i(1, 0, 0) > 0$;
3. $\mu_i(0, 1, 1) - \mu_i(0, 0, 1) < 0$;
4. $\mu_i(0, 1, 0) - \mu_i(0, 0, 0) < 0$.

- *Foreign comparisons* ($\mu_i(a_i, s, 1) - \mu_i(a_i, s, 0)$):

1. $\mu_i(1, 1, 1) - \mu_i(1, 1, 0) = 0$;

2. $\mu_i(0, 1, 1) - \mu_i(0, 1, 0) = 0$;
3. $\mu_i(1, 0, 1) - \mu_i(1, 0, 0) > 0$;
4. $\mu_i(0, 0, 1) - \mu_i(0, 0, 0) < 0$.

Proof of Remark A.1: I proceed in the order of cases enumerated in the remark.

- Domestic comparisons:

1. $\mu_i(1, 1, 1) = 1 > \mu_i(0, 1, 1) = 0$;
2. $\mu_i(1, 1, 0) = 1 > \mu_i(0, 1, 0) = 0$;
3. Immediate that if $\pi > \pi^\dagger$, $\mu_i(1, 0, 1) - \mu_i(0, 0, 1) < 0$. But if $\pi < \pi^\dagger$, $\mu_i(1, 0, 1) - \mu_i(0, 0, 1) > 0 \Leftrightarrow \pi^2 p_{j1}^2 (1 - G(\tilde{x}^*; 1)) G(\tilde{x}^*; 1) > (1 - \pi)^2 p_{j0}^2 (1 - G(\tilde{x}^*; 0)) G(\tilde{x}^*; 0)$ where we also use that $\beta^* = 1$. Rearranging gives $\frac{\pi}{1 - \pi} > \sqrt{\frac{G(\tilde{x}^*; 0)(1 - G(\tilde{x}^*; 0))^3}{G(\tilde{x}^*; 1)(1 - G(\tilde{x}^*; 1))^3}}$. By FOSD, $1 - G(\tilde{x}^*; 0) < 1 - G(\tilde{x}^*; 1)$, which guarantees the RHS is strictly less than the baseline threshold $\sqrt{\frac{G(\tilde{x}^*; 0)(1 - G(\tilde{x}^*; 0))}{G(\tilde{x}^*; 1)(1 - G(\tilde{x}^*; 1))}}$ that defines $\pi^\dagger$. So there is a cutoff $\pi^{\dagger\dagger} < \pi^\dagger$ such that $\mu_i(1, 0, 1) < \mu_i(0, 0, 1)$ if $\pi < \pi^{\dagger\dagger}$ and $\mu_i(1, 0, 1) > \mu_i(0, 0, 1)$ if $\pi^{\dagger\dagger} < \pi < \pi^\dagger$;
4. Immediate that if $\pi > \pi^\dagger$, $\mu_i(1, 0, 0) - \mu_i(0, 0, 0) < 0$. But if $\pi < \pi^\dagger$, $\mu_i(1, 0, 0) - \mu_i(0, 0, 0) > 0 \Leftrightarrow \pi^2 (1 - p_{j1}) G(\tilde{x}^*; 1)^2 (1 - G(\tilde{x}^*; 1)) > (1 - \pi)^2 (1 - p_{j0}) G(\tilde{x}^*; 0)^2 (1 - G(\tilde{x}^*; 0))$ where we also use that $\beta^* = 1$. Rearranging gives $\frac{\pi}{1 - \pi} > \sqrt{\frac{G(\tilde{x}^*; 0)^3 (1 - G(\tilde{x}^*; 0))}{G(\tilde{x}^*; 1)^3 (1 - G(\tilde{x}^*; 1))}}$. By FOSD, $G(\tilde{x}^*; 0) > G(\tilde{x}^*; 1)$, which guarantees the RHS is strictly greater than the baseline threshold $\sqrt{\frac{G(\tilde{x}^*; 0)(1 - G(\tilde{x}^*; 0))}{G(\tilde{x}^*; 1)(1 - G(\tilde{x}^*; 1))}}$ that defines $\pi^\dagger$. Hence for $\pi < \pi^\dagger$ we also have $\mu_i(1, 0, 0) - \mu_i(0, 0, 0) < 0$.

- Signal comparisons:

1. $\mu_i(1, 1, 1) = 1 > \mu_i(1, 0, 1)$;
2. $\mu_i(1, 1, 0) = 1 > \mu_i(1, 0, 0)$;
3. $\mu_i(0, 1, 1) = 0 < \mu_i(0, 0, 1)$;
4. $\mu_i(0, 1, 0) = 0 < \mu_i(0, 0, 0)$.

- Foreign comparisons:

1. $\mu_i(1, 1, 1) = \mu_i(1, 1, 0) = 1$;
2. $\mu_i(0, 1, 1) = \mu_i(0, 1, 0) = 0$;
3. $\mu_i(1, 0, 1) - \mu_i(1, 0, 0) > 0 \Leftrightarrow G(\tilde{x}^*; 0) > G(\tilde{x}^*; 1)$, true by FOSD;
4. $\mu_i(0, 0, 1) - \mu_i(0, 0, 0) < 0 \Leftrightarrow G(\tilde{x}^*; 1) < G(\tilde{x}^*; 0)$, true by FOSD.

■

Model Extensions

Here I briefly sketch two model extensions: a pro-climate interest group and a revised model in which the politician can condition directly on s .

Pro-Climate Interest Group

Suppose that instead of a special interest group biased against climate action, the group that disseminates information to the voter is in favor of ambitious climate policies. Specifically, the group designs a signal $s \in \{0, 1\}$ according to the experiment

$$\begin{aligned} P(s = 0|\omega = 0) &= 1 - \gamma. & P(s = 1|\omega = 0) &= \gamma. \\ P(s = 0|\omega = 1) &= 0. & P(s = 1|\omega = 1) &= 1. \end{aligned}$$

The interest group therefore chooses the parameter $\gamma \in [0, 1]$. All of the analysis remains as before. I characterize the equilibrium of the domestic model.

Proposition A.1 *In the unique equilibrium of the model with a pro-climate interest group:*

- *the probability of climate action is inverse U-shaped in γ ;*
- *there exists a π^γ such that the special interest's optimal misreporting strategy is*

$$\gamma^*(\pi) = \begin{cases} 0 & \pi < \pi^\gamma \\ 1 & \pi \geq \pi^\gamma. \end{cases}$$

Proof of Proposition A.1: First I show that the game has a unique equilibrium in the exact fashion as those in the main text. Then I demonstrate that $\tilde{x}^*$ is inverse U-shaped in γ . Finally I show that the interest group chooses $\gamma^* = 0$ when $\pi < \pi^\gamma$ and $\gamma^* = 1$ when $\pi > \pi^\gamma$.

Lemma A.15 *The model with a pro-climate interest group has a unique equilibrium given by the threshold $\tilde{x}^*$.*

Proof of Lemma A.15: The voter observes (a, s) and retains the politician when $\mu(a, s) \geq \varepsilon$, occurring with probability $F(\mu(a, s))$.

Given the signal x , and using similar notation as in the main text, the politician chooses $a = 1$ iff

$$2\eta(x) - 1 + (\eta(x) + (1 - \eta(x))\gamma)\Delta(1) + (1 - \eta(x))(1 - \gamma)\Delta(0) \geq 0.$$

Let $\tilde{x}$ be the value of x that solves this at equality. The posterior beliefs induced by these strategies are

$$\mu(1, 0) = 0.$$

$$\begin{aligned}\mu(1, 1) &= \frac{\pi(1 - G(\tilde{x}; 1))}{\pi(1 - G(\tilde{x}; 1)) + (1 - \pi)\gamma(1 - G(\tilde{x}; 0))}. \\ \mu(0, 0) &= 1. \\ \mu(0, 1) &= \frac{(1 - \pi)\gamma G(\tilde{x}; 0)}{(1 - \pi)\gamma G(\tilde{x}; 0) + \pi G(\tilde{x}; 1)}.\end{aligned}$$

Differentiating with respect to x yields

$$\frac{\partial \eta(\tilde{x})}{\partial \tilde{x}} \left(2 + (1 - \gamma)(\Delta(1; \tilde{x}) - \Delta(0; \tilde{x})) \right) + (\gamma(1 - \eta(\tilde{x})) + \eta(\tilde{x})) \frac{\partial \Delta(1; \tilde{x})}{\partial \tilde{x}}.$$

The first term is positive because $\eta(\tilde{x})$ is increasing in $\tilde{x}$ by MLRP. Since $\mu(1, 1; \tilde{x})$ is increasing in $\tilde{x}$ and $\mu(0, 1; \tilde{x})$ is decreasing in $\tilde{x}$, $\Delta(1; \tilde{x})$ is increasing in $\tilde{x}$. Then there is a $\tilde{x}^*$ such that the politician chooses $a = 1$ iff $x \geq \tilde{x}^*$. ■

Lemma A.16 *The equilibrium signal cutoff $\tilde{x}$ is inverse U-shaped in γ in the model with a pro-climate interest group.*

Proof of Lemma A.16: Define the function

$$I_\gamma(\tilde{x}) := 2\eta(\tilde{x}) - 1 + (\gamma - \gamma\eta(\tilde{x}) + \eta(\tilde{x}))\Delta(1; \tilde{x}) + (1 - \gamma)(1 - \eta(\tilde{x}))\Delta(0; \tilde{x}).$$

Clearly, $I_\gamma(\tilde{x})$ is increasing in $\tilde{x}$ and the point $\tilde{x}^*$ is defined by $I_\gamma(\tilde{x}^*) = 0$. By the implicit function theorem,

$$\frac{d\tilde{x}^*}{d\gamma} = -\frac{\partial I_\gamma(\tilde{x})/\partial \gamma}{\partial I_\gamma(\tilde{x})/\partial \tilde{x}}.$$

Differentiating with respect to γ yields

$$\frac{\partial I_\gamma(\tilde{x})}{\partial \gamma} = (1 - \eta(\tilde{x}))(\Delta(1; \tilde{x}) - \Delta(0; \tilde{x})) + (\gamma - \gamma\eta(\tilde{x}) + \eta(\tilde{x})) \frac{\partial \Delta(1; \tilde{x})}{\partial \gamma}.$$

Now, $\mu(1, 1)$ is decreasing in γ and $\mu(1, 0)$ is increasing in γ so $\frac{\partial \Delta(1; \tilde{x})}{\partial \gamma} < 0$. Then the first term is positive and the second term is negative. Analogous to Lemma A.6, this function is negative for high γ and positive for low γ . Hence by the implicit function theorem $\tilde{x}^*$ is U-shaped in γ . ■

Lemma A.17 *There exists π^γ such that $\gamma^* = 1$ when $\pi > \pi^\gamma$ and $\gamma^* = 0$ when $\pi < \pi^\gamma$.*

Proof of Lemma A.17: The argument is analogous to Lemma A.7 and so I omit details. As $\tilde{x}^*$ is U-shaped in γ , the probability of climate action is inverse U-shaped in γ . The special interest group seeks to maximize climate action, which happens at either $\gamma = 0$ or $\gamma = 1$. Since high π aligns with the interest group's preferred action, it selects $\gamma^* = 1$ when $\pi > \pi^\gamma$ and $\gamma^* = 0$ when $\pi < \pi^\gamma$. ■ ■

Politician and Interest Group Signal

In the main model, the politician is unable to condition her strategy on the signal s sent by the interest group. I now relax that assumption. This means that the politician's strategy is now a function her private signal x as well as the public signal s .

Proposition A.2 *In the model where the politician can observe the interest group's signal:*

- *the probability of climate action is increasing in β ;*
- *the optimal level of misreporting is $\beta^* = 0$.*

The results of this proposition directly illustrate the role of the politician's pre-message commitment. If the politician can observe the interest group's signal and the interest group inflates the possibility of false negatives, then the politician and the voter rationally discount this. This entices the politician to pursue action more, as the voter uses the politician's action more heavily as an assessment of appropriateness. These results also imply that there is an importance of having the politician not know what the interest group will ultimately report to the public.

Proof of Proposition A.2: Define $\rho(x, s; \beta) = P(\omega = 1|x, s; \beta)$ to be the politician's posterior belief that $\omega = 1$ given her private signal x and the realization of the interest group's message s given β . Since $s = 1$ is a truthful message, $\rho(x, 1) = 1$ for any value of x . Hence in the subgame following $s = 1$, the politician chooses $a = 1$ iff

$$1 + F(\mu(1, 1)) \geq F(\mu(0, 1)),$$

which is always satisfied: the politician always plays $a = 1$.

Following $s = 0$, the politician does not know if the special interest is truthfully reporting $\omega = 0$ or if with some probability β it misreported. Her posterior belief is $\rho(x, 0; \beta) = \frac{g(x;1)\beta\pi}{g(x;1)\beta\pi + g(x;0)(1-\pi)}$. Then the politician's problem is to choose $a = 1$ whenever

$$\rho(x, 0; \beta) + F(\mu(1, 0)) \geq (1 - \rho(x, 0; \beta)) + F(\mu(0, 0)) \Leftrightarrow 2\rho(x, 0; \beta) - 1 + \Delta(0) \geq 0.$$

The posterior beliefs $\mu(1, 0; \tilde{x})$ and $\mu(0, 0; \tilde{x})$ are defined as in Lemma A.4 and are thus increasing in $\tilde{x}$ and decreasing in $\tilde{x}$ respectively. Then there is a cutoff $\tilde{x}^*$ such that the politician plays $a = 1$ iff $x \geq \tilde{x}^*$.

Now observe that $\frac{\partial \rho(x, 0; \beta)}{\partial \beta} = \frac{\pi(1-\pi)g(x;1)g(x;0)}{(g(x;1)\pi\beta + g(x;0)(1-\pi))^2} > 0$ for any β and any signal x , as the politician rationally discounts the special interest group's signal as reporting is more likely to be biased. Moreover, as in Lemma A.6, $\frac{\partial \Delta(0)}{\partial \beta} > 0$ and so by the implicit function theorem $\frac{d\tilde{x}^*}{d\beta} \leq 0$, or misreporting increases climate action. This is simply because the politician rationally discounts the signal $s = 0$ at higher values of β . As action is monotonically increasing in β , the optimal misreporting is $\beta^* = 0$. ■

B Exxon Source Documents

This appendix provides more information on the documents described in Figure 4 of the main text that describe the trajectory of Exxon’s public messaging on the severity of climate change. They demonstrate how Exxon initially exploited uncertainty inherent to climate change, seeking to delay reforms by generating skepticism among the public. This is particularly apparent in the 1998 Victory Memo. However, by 2014, ExxonMobil acknowledged the risks of climate change.

- In 1996 and 1998, Exxon released pamphlets to the masses that sought to inject doubt into the public discourse about the validity of climate science and the subsequent need for policy action. In particular, one such pamphlet was entitled “Global climate change: everyone’s debate,” and is available at <https://www.climatefiles.com/exxonmobil/1998-exxon-pamphlet-global-climate-change-everyones-debate/>. The pamphlet “Global warming: who’s right?” admonishes readers not to “ignore the facts” about climate change and is available at <https://climateintegrity.org/uploads/deception/1996-Exxon-Global-Warming-Whos-Right.pdf>.
- The “Victory Memo” of 1998 makes the goal to inject uncertainty into the public sphere clear: “victory will be achieved when average citizens ‘understand’ (recognize) uncertainties in climate science,” and “recognition of uncertainty becomes part of the ‘conventional wisdom.’” The memo can be found at <https://www.climatefiles.com/trade-group/american-petroleum-institute/1998-global-climate-science-communications-team-action-plan/>.
- The Global Climate Coalition was a lobbying group of several large oil and gas companies that operated between 1989 and 2001. Its primary function was to coordinate messaging against global climate action like the ratification of the Kyoto Protocol. In 1995, the GCC internally circulated *Predicting Future Climate Change: A Primer*, which summarized the state of climate science. Notably, it reads, “The scientific basis for the Greenhouse Effect and the potential impact of human emissions of greenhouse gases such as CO₂ on climate is well established and cannot be denied.” The full document is available at https://www.ucsusa.org/sites/default/files/attach/2015/07/Climate-Deception-Dossier-7_GCC-Climate-Primer.pdf.
- While the GCC internally circulated *Predicting Future Climate Change: A Primer*, its public-facing publications of the time were very different. In 1995, it also published “Climate Change: Your Passport To The Facts,” a booklet allegedly intended to introduce readers to essential facts about climate change. Facts include that “the notion that scientists have reached consensus that man-made emissions of greenhouse gases are leading to a dangerous level of global warming is not true” and “computer climate models, which are the basis for ”predictions” of global climate change, suffer from severe flaws.” The document is available at <https://www.worthingtoncaron.com/documents/1995-CLIMATE-CHANGE-YOUR-PASSPORT.pdf>.

- ExxonMobil published a series of newspaper ads in order to sow doubt into the public about climate science. In the spring of 2000, ExxonMobil ran the ad “Unsettled Science” in major news outlets (e.g., the *New York Times*). These ads also tried to discredit climate scientists, see <https://insideclimatenews.org/news/22102015/exxon-sowed-doubt-about-climate-science-for-decades-by-stressing-uncertainty/>.
- ExxonMobil was known to fund several European think tanks to discourage climate reforms in countries beyond the United States. See <https://archive.corporateeurope.org/longerversioncleaningtheair.html>.

C Empirical Implications

I study empirical implications of the model using data from Eurobarometer and the Climate Change Laws of the World Project. Across seven survey waves (2011, 2013, 2015, 2017, 2019, 2021, 2023) administered in 27 EU countries, Eurobarometer asks: “how serious a problem do you think climate change is at this moment? Please use a scale from 1 to 10, ‘1’ would mean that it is ‘not at all a serious problem’ and ‘10’ would mean that it is a problem that is ‘extremely serious.’ ” I take the average of respondents’ answers within each country for each wave such that I have a measure of average beliefs at the country-year level.

The CCLW data provides information about climate law adoption for 194 countries. I code a simple binary indicator that equals 1 if a country adopts a climate law in a given year and is 0 otherwise. I subset the data to laws that are related to mitigation, and also remove any documents in the dataset flagged as “UNFCCC submissions” (these are just repositories of countries’ Nationally Determined Contributions from the Paris Agreement) to study implemented laws.

I construct a dyadic panel to most easily study the relationship between domestic and foreign law adoption and climate beliefs. Each dyad is thus described by columns `iso1`, `iso2`, and `year`. Within each dyad, I let the country designated as `iso2` be the home (European only for which we have public opinion data) country i , and the country designated as `iso1` be the foreign (global) country j . Table A.1 reports descriptive statistics for the variables used in the analysis.

Statistic	Mean	St. Dev.	Min	Max
<i>Panel A: Full Sample ($N = 37,222$)</i>				
Avg. Belief CC Serious	7.563	0.626	5.587	8.852
Climate Law Adoption (Home)	0.710	0.454	0	1
Climate Law Adoption (Foreign)	0.502	0.500	0	1
<i>Panel B: Pre-2014, Exxon = 0 ($N = 10,588$)</i>				
Avg. Belief CC Serious	7.338	0.674	5.587	8.852
Climate Law Adoption (Home)	0.546	0.498	0	1
Climate Law Adoption (Foreign)	0.440	0.496	0	1
<i>Panel C: Post-2014, Exxon = 1 ($N = 26,634$)</i>				
Avg. Belief CC Serious	7.652	0.582	5.676	8.757
Climate Law Adoption (Home)	0.775	0.417	0	1
Climate Law Adoption (Foreign)	0.527	0.499	0	1

Table A.1: Descriptive Statistics

The theory defines the voter’s posterior belief about climate policy appropriateness based on three factors: domestic climate action a_i , special interest messaging s , and foreign climate action a_j . Given the dyadic structure of the data, a dyad-year is thus completely characterized by one of eight combinations of these three indicators.

I first regress average seriousness on dyad and year dummies to demean the dependent variable, and then regress the residuals of that variable on indicators for each of the eight cells in a second stage (equivalent to dyad and year fixed effects). Table A.2 provides the cell means shown in Figure 5 of the main text. Observations are demeaned by dyad and year prior to aggregation into the cells.

a_i	s	a_j	Avg. Belief CC Serious	N
0	0	0	0.0783	2697
0	0	1	0.0516	2115
1	0	0	-0.0588	3236
1	0	1	-0.0512	2540
0	1	0	-0.0731	2852
0	1	1	-0.0778	3131
1	1	0	0.0248	9739
1	1	1	0.0193	10912

Table A.2: Average Beliefs by Triple

These means correspond to $E[CCSerious|a_i, s, a_j]$ for different permutations of a_i , s , and a_j . The theory, fully elaborated in Remark A.1, gives us leverage on the sign of differences of these beliefs. For example, if we vary domestic climate action, the theory can sign

$$E[CCSerious|a_i = 1, s, a_j] - E[CCSerious|a_i = 0, s, a_j],$$

which is simply a linear hypothesis test using the cell means—the regression coefficients of the second stage regression described above.

The coefficients of this second stage regression are used to form the linear hypothesis tests reported in Figure 6 and Table A.3.

Comparison	Estimate
<i>Domestic Comparisons: $E[CC\ Serious a_i = 1, s, a_j] - E[CC\ Serious a_i = 0, s, a_j]$</i>	
$s = 1, a_j = 1$	0.097*** (0.006)
$s = 0, a_j = 1$	-0.103*** (0.010)
$s = 1, a_j = 0$	0.098*** (0.005)
$s = 0, a_j = 0$	-0.137*** (0.009)
<i>Signal Comparisons: $E[CC\ Serious a_i, s = 1, a_j] - E[CC\ Serious a_i, s = 0, a_j]$</i>	
$a_i = 1, a_j = 1$	0.071*** (0.008)
$a_i = 0, a_j = 1$	-0.129*** (0.011)
$a_i = 1, a_j = 0$	0.084*** (0.008)
$a_i = 0, a_j = 0$	-0.151*** (0.010)
<i>Foreign Comparisons: $E[CC\ Serious a_i, s, a_j = 1] - E[CC\ Serious a_i, s, a_j = 0]$</i>	
$a_i = 1, s = 1$	-0.005 (0.004)
$a_i = 0, s = 1$	-0.005 (0.007)
$a_i = 1, s = 0$	0.008 (0.009)
$a_i = 0, s = 0$	-0.027** (0.012)

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the dyad level

Table A.3: Linear Hypothesis Tests

Because a climate law adopted late in a calendar year may postdate the fielding of that year's survey wave, the main specification lags adoption by one year to preserve temporal ordering between the dependent and independent variables. Table A.4 reports the results of the linear hypothesis tests if climate adoption is not lagged. Results are directionally consistent with those in the main text, though the domestic contrast for $s = 1, a_j = 0$ is no longer statistically significant at conventional levels.

Comparison	Estimate
<i>Domestic Comparisons: $E[CC\ Serious a_i = 1, s, a_j] - E[CC\ Serious a_i = 0, s, a_j]$</i>	
$s = 1, a_j = 1$	0.014** (0.006)
$s = 0, a_j = 1$	-0.170*** (0.011)
$s = 1, a_j = 0$	0.007 (0.006)
$s = 0, a_j = 0$	-0.178*** (0.010)
<i>Signal Comparisons: $E[CC\ Serious a_i, s = 1, a_j] - E[CC\ Serious a_i, s = 0, a_j]$</i>	
$a_i = 1, a_j = 1$	0.081*** (0.009)
$a_i = 0, a_j = 1$	-0.102*** (0.010)
$a_i = 1, a_j = 0$	0.071*** (0.009)
$a_i = 0, a_j = 0$	-0.114*** (0.009)
<i>Foreign Comparisons: $E[CC\ Serious a_i, s, a_j = 1] - E[CC\ Serious a_i, s, a_j = 0]$</i>	
$a_i = 1, s = 1$	-0.000 (0.004)
$a_i = 0, s = 1$	-0.007 (0.007)
$a_i = 1, s = 0$	-0.011 (0.010)
$a_i = 0, s = 0$	-0.019* (0.011)

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the dyad level

Table A.4: Linear Hypothesis Tests (No Lagged Climate Adoption)